Description

Image





Caption

1. Filament of an incandescent lamp in pure tungsten. © ANSYS, Inc. 2. Spark plug with tungsten electrodes, having to survive in the combustion chamber and tolerate spark discharges. © ANSYS, Inc.

The material

Tungsten is remarkable for its melting point of 3410 °C when pure. This is a problem if you want to process it, but an attraction if you want to use it at high temperatures. Processing is solved by using powder methods -- the same methods used to shape refractory ceramics. Its high melting point gives tungsten excellent creep resistance up to 1400 °C (the temperature listed below as the maximum service temperature), but if not under load it can be used at much higher temperatures: photoflood bulbs reach 2200 °C, though they have a short life.

Compositional summary

> 99% W

General properties

Density		1.69e4	-	1.86e4	kg/m^3
Price	*	28.3	-	50.3	GBP/kg
Date first used		1783			

Mechanical properties

Young's modulus		310	-	370	GPa
Shear modulus		120	-	140	GPa
Bulk modulus		220	-	290	GPa
Poisson's ratio		0.27	-	0.29	
Yield strength (elastic limit)		490	-	1.23e3	MPa
Tensile strength		751	-	1.44e3	MPa
Compressive strength	*	523	-	1.23e3	MPa

Elongation		1.2	-	19	% strain
Hardness - Vickers		268	-	419	HV
Fatigue strength at 10^7 cycles	*	279	-	593	MPa
Fracture toughness	*	120	-	150	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	1e-4	-	3e-4	

Thermal properties

Melting point	*	3e3	-	3.3e3	°C
Maximum service temperature	*	817	-	957	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		71.3	-	116	W/m.°C
Specific heat capacity	*	130	-	140	J/kg.°C
Thermal expansion coefficient		4.26	-	5.7	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		9.3	-	14	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		1			
Formability		1	-	2	
Machinability		1	-	2	
Weldability		3	-	4	
Solder/brazability		1	-	2	

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Acceptable			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Acceptable			

Durability: acids

Acetic acid (10%)	Acceptable
Acetic acid (glacial)	Acceptable
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Limited use
Nitric acid (70%)	Limited use
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Limited use
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Acceptable
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent

Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Limited use
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Acceptable
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Excellent
Tolerance above 850 C (1562 F)	Excellent

Geo-economic data for principal component

Annual world production, principal component	7.6e4	tonne/yr
Reserves, principal component	3.06e6	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 36.2	-	39.9	kg/kg
Embodied energy (virgin grade production)	* 550	-	606	MJ/kg
Water usage	* 147	-	162	l/kg

Material processing: energy

Metal casting, embodied energy	* 42.3	-	47.3	MJ/kg
Metal extrusion, hot, embodied energy	* 28.1	-	38.1	MJ/kg
Metal rolling and forging, embodied energy	* 14.6	-	19.7	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg

Metal powder forming, embodied energy	* 47.3	-	54.6	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 7.04e3	-	7.99e3	MJ/kg
Machining, coarse, embodied energy	* 18.7	-	21.7	MJ/kg
Machining, fine, embodied energy	* 26.9	-	32.9	MJ/kg
Machining, grinding, embodied energy	* 35.1	-	44.1	MJ/kg
Machining, laser, CO2, embodied energy	* 0.838	-	0.951	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.99	-	3.35	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.84	-	2.5	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.955	-	1.29	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 3.85	-	4.44	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 579	-	657	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.22	-	1.42	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.76	-	2.15	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.29	-	2.89	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.069	-	0.0782	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 21.4	-	23.5	kg/kg
Embodied energy, recycling	* 286	-	315	MJ/kg
Recycle fraction in current supply	46			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Tungsten is a skin and eye irritant and is noxious if ingested.

Supporting information

Design guidelines

Tungsten is heavy, relatively expensive and difficult to process -- it is a choice of last-resort. Most of its applications exploit its very high melting point (spark plug electrodes and lamp filament), its exceptional tensile strength when drawn to wire (reinforcement in metal, ceramic and polymer-matrix composites) or it very high density (armor-piercing penetrators). Because of the difficulty of processing tungsten it is generally available only as wire, rod or sheet.

Technical notes

Tungsten is produced by a chemical route that delivers the metal as a fine powder. The powder is pressed into billets, sintered and swaged to rod, then drawn to wire or rolled to sheet. Wires as fine as 5 microns in diameter are available. They can be woven into fabric or used as reinforcement in other metals, ceramics or polymers.

Typical uses

Applications are of four types.
1. Those using the high-temperature capability of tungsten: spark-plug electrodes, lamp filaments, heating elements and furnace windings and electrodes for TIG welding.
2. Those using the high density: balance weights, anti-vibration tooling, armor-piercing penetrators and radiation shielding, and X and gamma-ray shielding
3. Those using the high strength: reinforcement in composites, surface coatings for abrasion resistance
4. Those using its ability to harden steel: tool steels and armor.

Links

ProcessUniverse

Producers

Reference